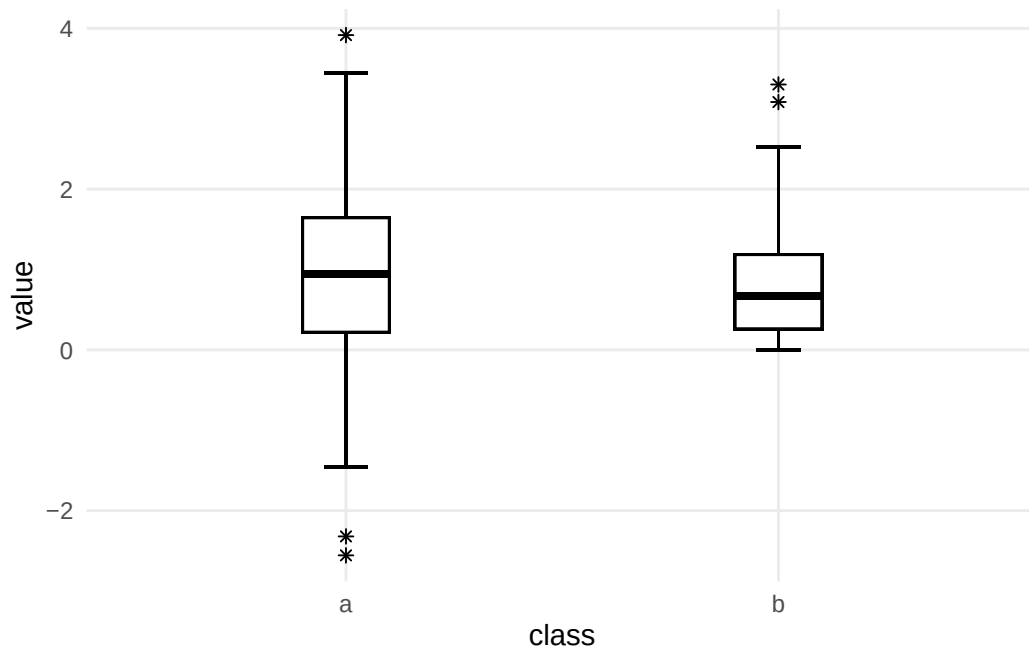


25-10-22_update

Update to make_template creating a file and opening into new window

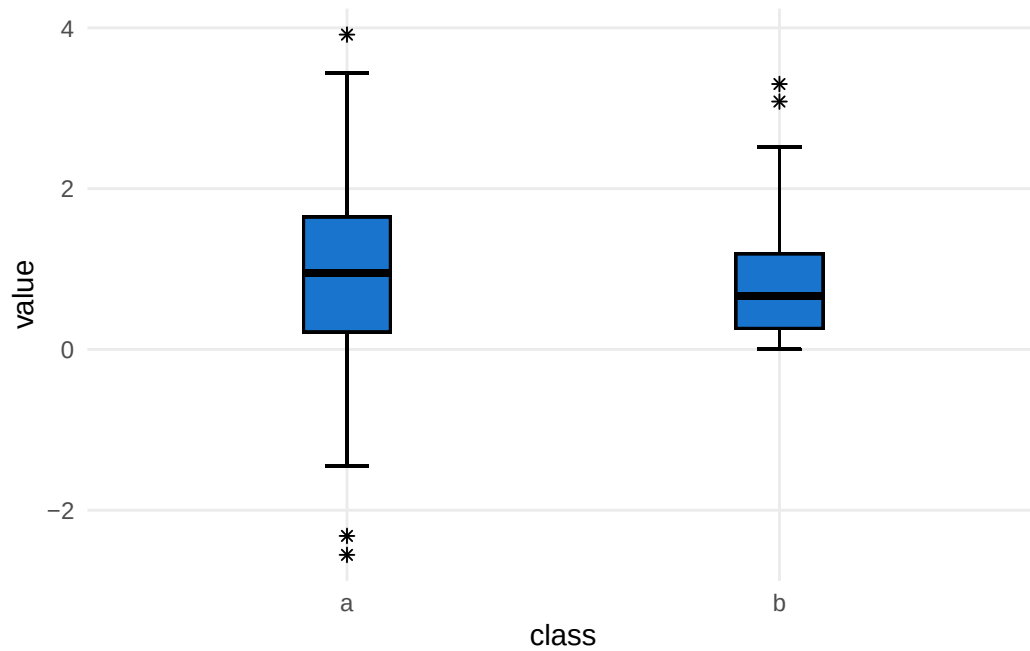
```
#Make Data  
set.seed(17)  
a <- tibble(value = c(rnorm(2e2, 1), 4*rbeta(2e2, 1, 4)),  
             class = c(rep("a", 2e2), rep("b", 2e2)))  
boxplot_template(a)
```



```
#Create Blue template that has a blue fill
#make_template("boxplot", "blue")
read_lines(file = "blue.R")
```

```
[1] "#Could concievably put an explanation of how to edit the code here"
[2] ""
[3] "x <- x |>"
[4] "  group_by(class) |>"
[5] "  mutate("
[6] "    q1 = quantile(value, 0.25),"
[7] "    q3 = quantile(value, 0.75),"
[8] "    median = median(value),"
[9] "    iqr = q3 - q1,"
[10] "    lower = q1 - 1.5 * iqr,"
[11] "    upper = q3 + 1.5 * iqr,"
[12] "    outlier = value < lower | value > upper,"
[13] "    min = min(ifelse(!outlier, value, NA), na.rm = TRUE),"
[14] "    max = max(ifelse(!outlier, value, NA), na.rm = TRUE)"
[15] "  ) |>"
[16] "  ungroup()"
[17] ""
[18] "print("
[19] "  ggplot(x, aes(x = class, y = value))+
[20] "    geom_point(data = x |> filter(outlier == TRUE), pch = 8)+
[21] "    geom_errorbar(aes(ymin = min, ymax = max), width = 0.1)+
[22] "    geom_crossbar(aes(ymin = q1, y = median, ymax = q3),"
[23] "                  width = 0.2, fill = \"dodgerblue3\")"
[24] "  )"
```

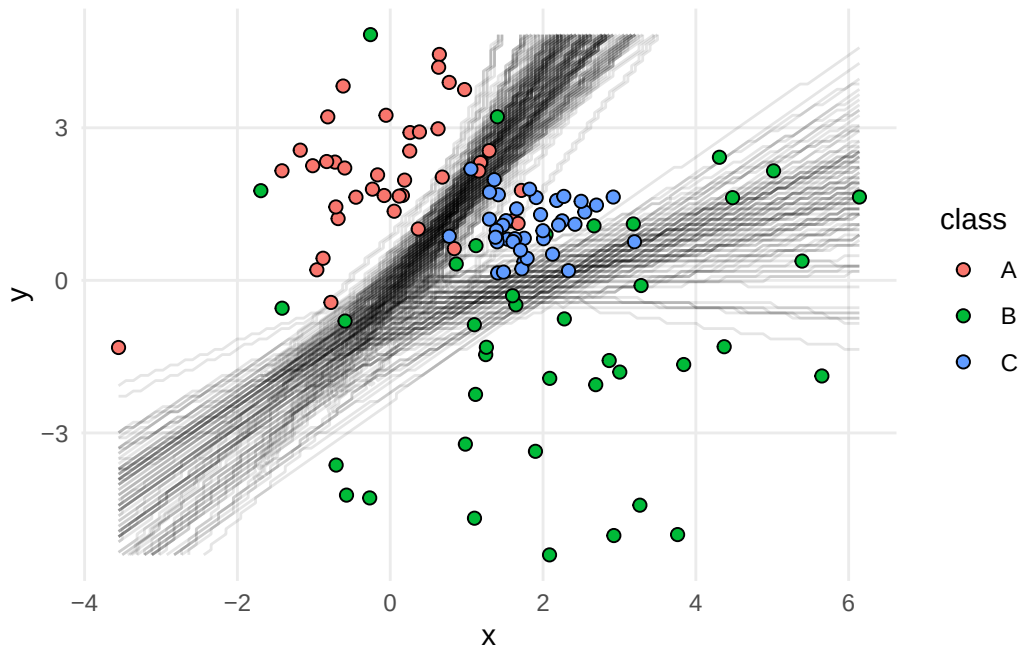
```
#make the boxplot with the new template
boxplot_template(a, "blue.R")
```



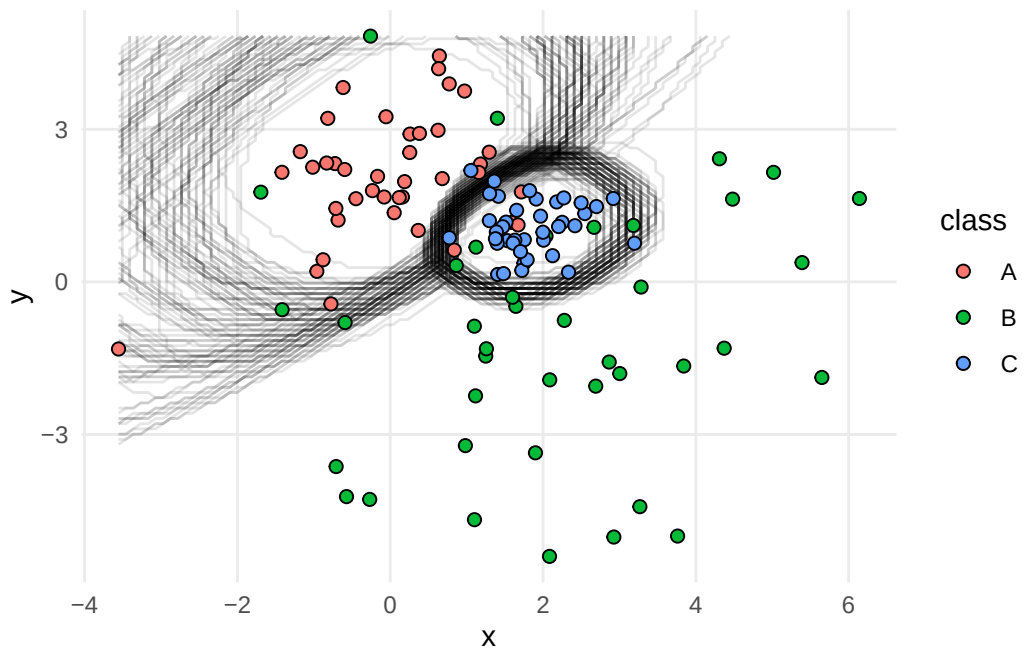
Create wrapper functions in ggclassify

Boundary bootwrapping function

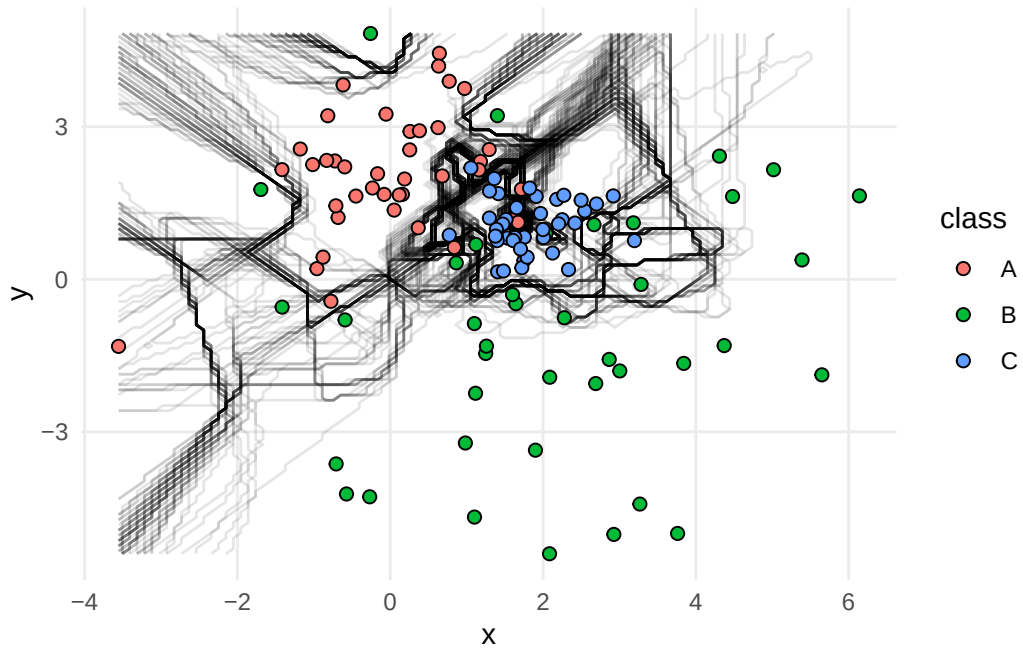
```
simpleplot <- ggplot(df, aes(x, y, fill = class))+  
  geom_point_fill()  
  
classify_boundary_boot(df, simpleplot, n = 100, method = "lda")
```



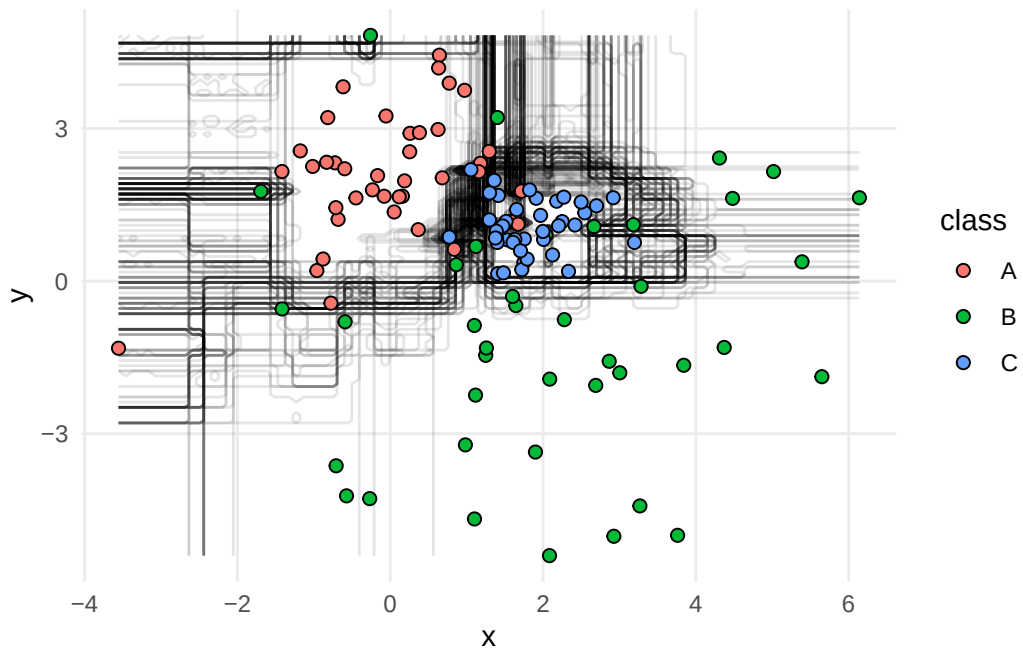
```
classify_boundary_boot(df, simpleplot, n = 100, method = "qda")
```



```
classify_boundary_boot(df, simpleplot, n = 100, method = "knn")
```



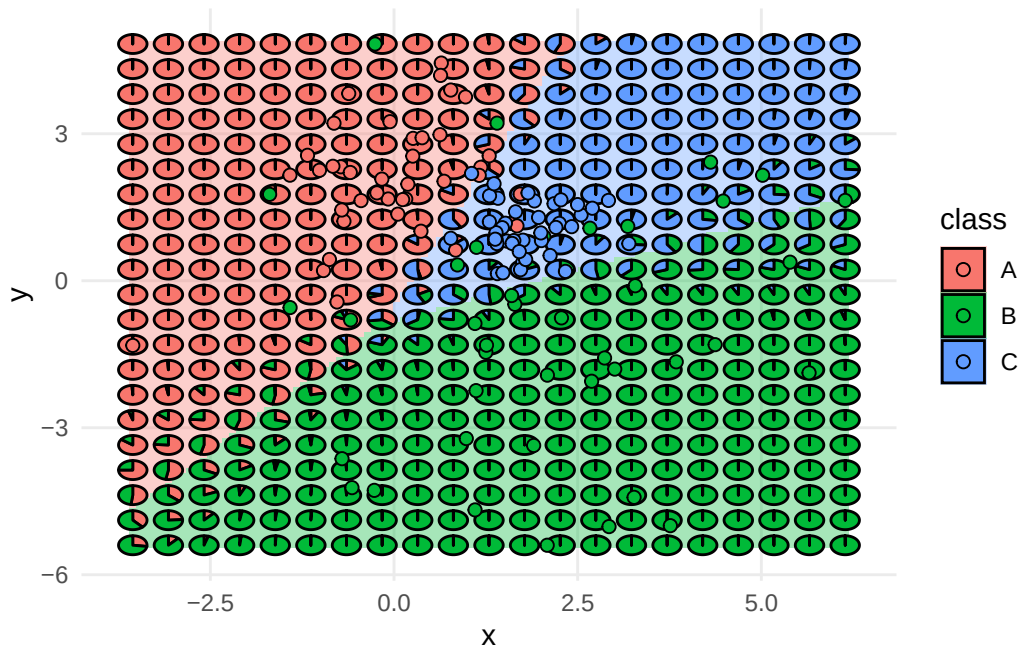
```
classify_boundary_boot(df, simpleplot, n = 100, method = "randomForest")
```



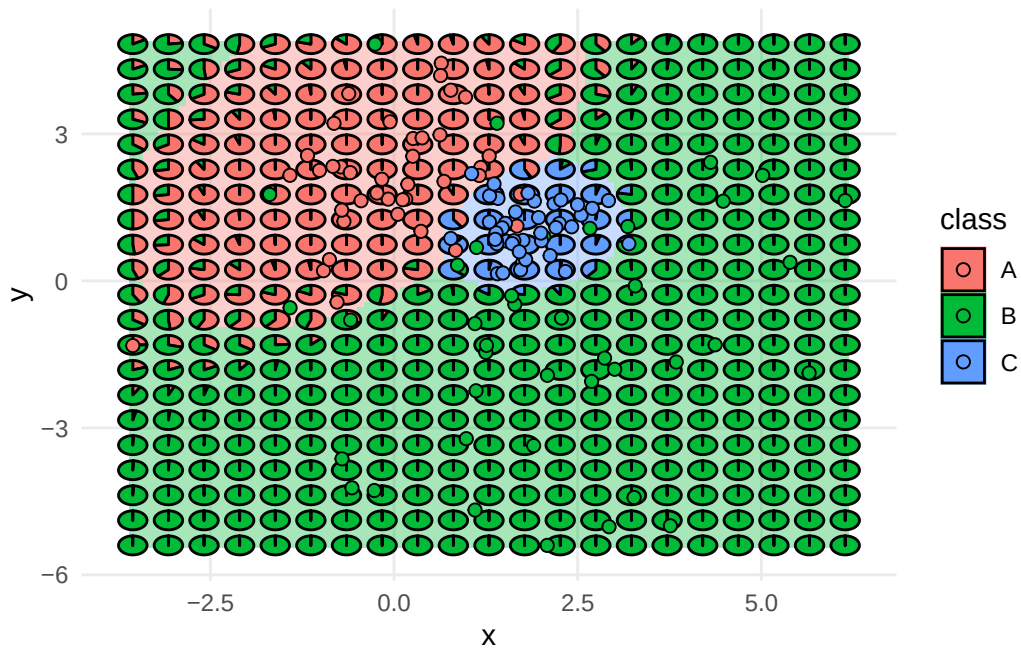
Piechart style bootwrapping function

The current issue is that this wrapper relies upon a geom from another package with multiple dependencies.

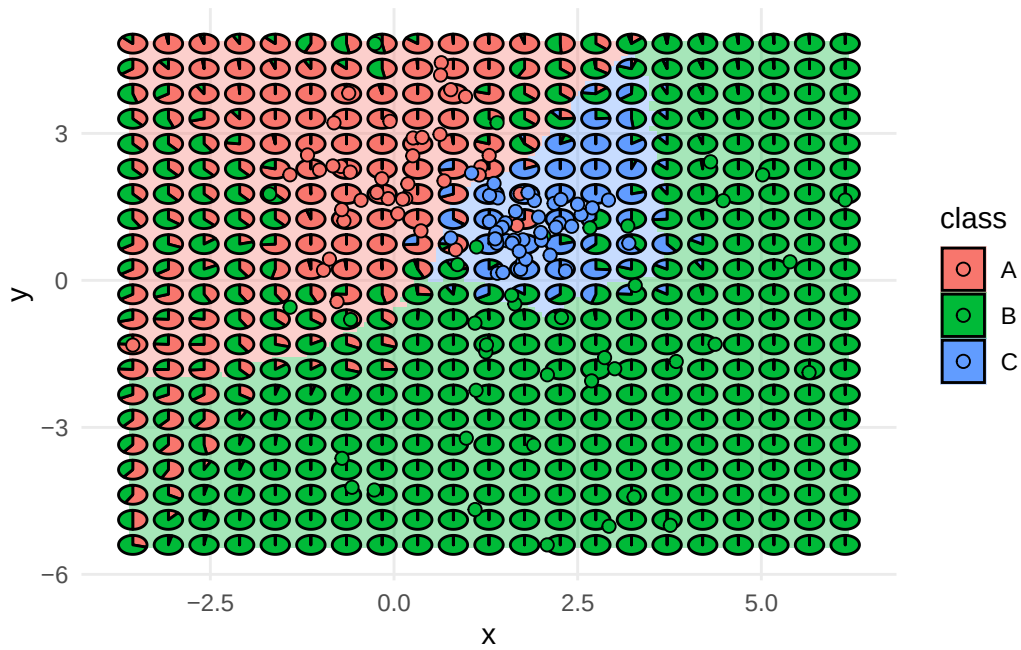
```
classify_boot(df, simpleplot, n = 100, res = 21, method = "lda")
```



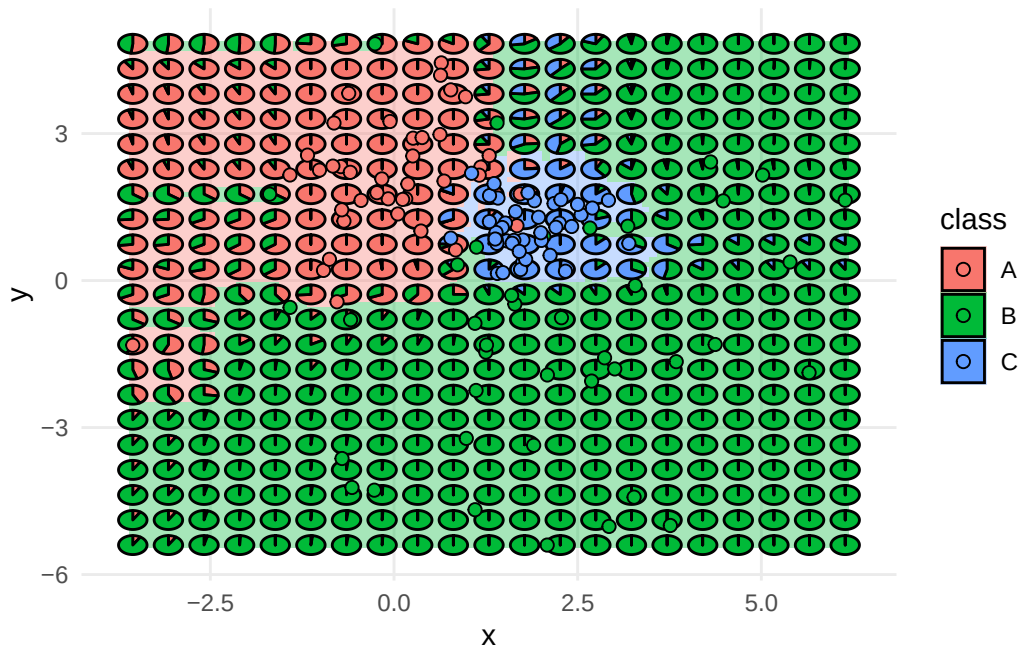
```
classify_boot(df, simpleplot, n = 100, res = 21, method = "qda")
```



```
classify_boot(df, simpleplot, n = 100, res = 21, method = "knn")
```



```
classify_boot(df, simpleplot, n = 100, res = 21, method = "randomForest")
```



Non-progressed

Geom_Cluster

Have not yet completed rewriting tidyclust implementation into separate geom_cluster

Potrace

Continuing work on the outline for wrapper function paper, looks something like this

```
# Wrapper Functions Background

## What are wrapper functions
### Functions that try to make a complex process easy to run
### ggplot wrapper functions (the focus of this paper?)

## Advantages to wrappers
### Complex analysis and plots can be created in one line by new user
### examples of popular wrappers

## How they can obscure user choice
### plot and function parameters that aren't adjustable
### example of wrapper plots that can be difficult to edit (time series comes to mind)
### Complex crash messages that aren't clear about their source

# Issues with developing wrapper functions

## The balance between simplicity for user, and ability to customize.
### More options for input can lead to increasing code complexity when trying to account for

## Self contained versus non self contained wrapper functions?
### Functions that call other functions
#### Functions that call non-exported functions
#### Functions that call C code.
### Functions that utilize S3 or other methods.

# Potential solutions to simplicity & Customization tradeoff

## "original" method (Perhaps so simple that I just combine it and the name method which have)
### The simplest default solution I originally developed
### Runs into issues if making more than 1 or 2 custom plots

## name method
### A slight improvement on that in the form of the "name" method
### Allows for storing multiple different customized plots since each has separate binding name

## binding method
### Can utilize S3 print methods and is more explicit
```

```

### Doesn't save lists into user global without their knowlege

## template method
### Uses .R files to store custom made templates for further use

## Why i think the template method is the best
### Templates can easily be shared between various users or stored between sessions
### Easier to further edit an R file after saving as opposed to a list stored in your global

# Utilizing these solutions on pre - existing wrappers

## Simple method for editing other functions
### basic function that can pull code of a function into a new window for editing and saving
### Will run into issues on any non-self contained wrapper function (many of them)
### The best implementations will involve the author including these methods of editing them

## Not sure but If i try to make a roxygen tag it goes here

## how wrapper writers could utilize these potential solutions (This might should be done in

```